Precision Determination of Gravitational Acceleration g via a Digital-Telemetry Reversible Compound Pendulum and Bessel's Asymmetric Formulation

Abstract

We report on the high-precision determination of the local gravitational acceleration *g* using a modern digital-telemetry reversible compound pendulum (Kater's pendulum). Rather than relying on classical optical cathetometers or manually triggered stopwatches, our apparatus integrates an embedded 6-degree-of-freedom inertial measurement unit (MPU-6050) sampling at 100 Hz, streaming real-time angular velocity and tri-axial acceleration. Using sub-sample linear interpolation at gyroscope zero-crossings, half-swing numerical velocity integration ($\theta_0 = \frac{1}{2}\int |\omega| dt$), and per-cycle analytic Borda finite-amplitude circular corrections, we evaluate two complete multi-trial datasets gathered across independent laboratory sessions.

In Dataset 1 (October 15, 2026), across $N_1 = 49$ and $N_2 = 52$ oscillations, we obtain corrected periods of $T_{1,0} = 1.69042 \pm 0.00090$ s (sample SD, standard error of the mean $\sigma_{\bar{T}} = 0.128$ ms) and $T_{2,0} = 1.77322 \pm 0.00144$ s ($\sigma_{\bar{T}} = 0.200$ ms). Applying Bessel's asymmetric compound pendulum formula for knife-edge separation $L = 1.0000 \pm 0.0005$ m and center-of-mass balance asymmetry $h_1 - h_2 = -0.1400 \pm 0.0014$ m yields $g = 9.8079 \pm 0.0264\ \mathrm{m/s^2}$ ($\pm 0.27\%$ total analytical uncertainty), in agreement with conventional standard gravity $g_0 = 9.80665\ \mathrm{m/s^2}$ ($+0.013\%$ relative discrepancy, well within the experimental uncertainty budget). In Dataset 2 (October 17, 2026), across $N_1 = 39$ and $N_2 = 105$ oscillations, we obtain $T_{1,0} = 1.68440 \pm 0.00160$ s ($\sigma_{\bar{T}} = 0.257$ ms) and $T_{2,0} = 1.77187 \pm 0.00706$ s ($\sigma_{\bar{T}} = 0.689$ ms), yielding $g = 9.7045 \pm 0.0381\ \mathrm{m/s^2}$ (-1.04% relative discrepancy).

Furthermore, we provide an engineering post-mortem explaining the origins of three initial anomalies observed during live testing: (i) the naive symmetric formula artifact producing $g \approx 13.2$--$13.8\ \mathrm{m/s^2}$, (ii) the low-amplitude angular speed cutoff bug that generated missed zero crossings and collapsed Bessel's calculation to $g \approx 4.81\ \mathrm{m/s^2}$ (-50.87%), and (iii) the mounting tilt asymmetry that caused decay-plot splitting. The physical telemetry from both sessions is demonstrated to be experimentally robust, repeatable, and capable of sub-percent precision ($\pm 0.27\%$).

1. Introduction & Historical Context

The determination of the acceleration due to terrestrial gravity, *g*, has historically occupied a foundational role in geodesy, metrology, and classical mechanics. In 1818, Captain Henry Kater designed the reversible compound pendulum to overcome the principal obstacle plaguing simple string pendulums: the impossibility of locating the exact center of oscillation of a finite extended mass [1]. By invoking Christiaan Huygens's dynamical principle of reciprocity—stipulating that the center of suspension and center of oscillation are mutually interchangeable—Kater proved that if a rigid bar oscillates with identical periods $T_1 = T_2 = T$ about two opposing knife edges, the distance L between the edges corresponds precisely to the equivalent length of a simple pendulum:

$$g = \frac{4\pi^2 L}{T^2}$$

However, mechanically adjusting movable brass counterweights until T_1 and T_2 are identical to within microsecond precision is tedious and sensitive to ambient thermal and mechanical drifts. To eliminate this operational bottleneck, Friedrich Wilhelm Bessel developed an exact analytical formulation in 1828 for compound pendulums with unequal periods [2]. Bessel demonstrated that by incorporating the asymmetry of the center of mass relative to the two knife edges $(h_1 - h_2)$, one can compute g directly from two non-identical periods $T_1 \neq T_2$ without requiring mechanical period matching.

In this work, we modernize the classical Kater–Bessel experiment by coupling a rigid reversible bar to a digital inertial measurement telemetry pipeline. By streaming angular velocity $\omega_z(t)$ at 100 Hz from an embedded Micro-Electro-Mechanical Systems (MEMS) sensor, we achieve high statistical power over hundreds of cycles, eliminate human reaction time errors, record full amplitude decay envelopes, and perform rigorous propagation of uncertainty.

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2. Theoretical Principles

2.1 The Compound Pendulum & Center of Oscillation

Consider a rigid body of mass M pivoted about a horizontal knife edge K_1 located at a distance h_1 from its center of mass (CoM). By the parallel axis theorem, the moment of inertia about K_1 is:

$$I_1 = I_{\text{cm}} + Mh_1^2 = M(k^2 + h_1^2)$$

where k is the radius of gyration about the center of mass. For small angular displacements $\theta \ll 1$, the equation of motion is:

$$I_1\ddot{\theta} + Mgh_1\theta = 0$$

yielding the natural period of oscillation:

$$T_1 = 2\pi\sqrt{\frac{I_1}{Mgh_1}} = 2\pi\sqrt{\frac{k^2 + h_1^2}{gh_1}}$$

Squaring both sides and rearranging:

$$\frac{gh_1T_1^2}{4\pi^2} = k^2 + h_1^2 \implies \frac{g}{4\pi^2}h_1T_1^2 - h_1^2 = k^2 \tag{1}$$

Now, let the pendulum be inverted and suspended from the second knife edge K_2, located on the opposite side of the center of mass at a distance h_2. The corresponding period is:

$$T_2 = 2\pi\sqrt{\frac{k^2 + h_2^2}{gh_2}} \implies \frac{g}{4\pi^2}h_2T_2^2 - h_2^2 = k^2 \tag{2}$$

Equating (1) and (2) eliminates the unknown moment of inertia k^2:

$$\frac{g}{4\pi^2}\left(h_1T_1^2 - h_2T_2^2\right) = h_1^2 - h_2^2 = (h_1 + h_2)(h_1 - h_2) \tag{3}$$

2.2 Bessel's Asymmetric Formulation

Let $L = h_1 + h_2$ denote the total physical distance between the two knife edges. We can express h_1 and h_2 in terms of L and their difference $(h_1 - h_2)$:

$$h_1 = \frac{L + (h_1 - h_2)}{2}, \quad h_2 = \frac{L - (h_1 - h_2)}{2}$$

Substituting these expressions into the left side of Eq. (3):

$$\begin{aligned} h_1 T_1^2 - h_2 T_2^2 &= \frac{L + (h_1 - h_2)}{2} T_1^2 - \frac{L - (h_1 - h_2)}{2} T_2^2 \\ &= \frac{L}{2}(T_1^2 - T_2^2) + \frac{h_1 - h_2}{2}(T_1^2 + T_2^2) \end{aligned}$$

Dividing Eq. (3) through by $(h_1 + h_2)(h_1 - h_2) = L(h_1 - h_2)$:

$$\frac{g}{4\pi^2}\left[\frac{T_1^2 + T_2^2}{2L} + \frac{T_1^2 - T_2^2}{2(h_1 - h_2)}\right] = 1$$

Multiplying through by 2 and solving for *g* yields **Bessel's exact formula**:

$$g = \frac{8\pi^2}{\dfrac{T_1^2 + T_2^2}{L} + \dfrac{T_1^2 - T_2^2}{h_1 - h_2}} \tag{4}$$

Mathematical Significance of the Asymmetric Formulation

Bessel's formula possesses a significant error-attenuating property: * In the denominator, the dominant term is $\frac{T_1^2+T_2^2}{L}$. The knife-edge separation L can be measured with high accuracy using a precision vernier cathetometer ($\Delta L \approx \pm 0.5$ mm, or $\pm 0.05\%$). * The second term, $\frac{T_1^2-T_2^2}{h_1-h_2}$, contains the center-of-mass balance asymmetry $(h_1 - h_2)$. Because finding the center of mass by knife-edge balance introduces an experimental uncertainty ($\Delta(h_1 - h_2) \approx \pm 1.4$ mm, or $\pm 1.0\%$), this term would normally inject large errors into *g*. * However, because T_1 and T_2 are close in value ($T_1 \approx 1.69$ s, $T_2 \approx 1.77$ s), the numerator $(T_1^2 - T_2^2)$ is small ($2.86 - 3.14 = -0.28$ s^2). * Defining the denominator sum $\Phi = \frac{T_1^2+T_2^2}{L} + \frac{T_1^2-T_2^2}{h_1-h_2}$, the relative sensitivity of *g* to $(h_1 - h_2) \equiv d$ is:

$$\frac{\delta g}{g} \approx \left(\frac{T_1^2 - T_2^2}{\Phi d}\right)\frac{\delta d}{d}$$

For our apparatus, $\Phi \approx 8.05$ s^2/m and $\frac{T_1^2-T_2^2}{d} \approx +2.05$ s^2/m, yielding a multiplier of $\frac{2.05}{8.05} \approx 0.25$. * Consequently, relative uncertainty in the center of mass is attenuated by a factor of approximately $4\times$: a 1.0% measurement uncertainty in $(h_1 - h_2)$ propagates to only a $\approx 0.25\%$ uncertainty in *g*. Because period errors are negligible over tens of cycles, the balance asymmetry measurement remains the primary contributor ($\approx 89\%$) to the total experimental error variance.

2.3 Finite-Amplitude (Borda) Circular Correction

The linearizing approximation $\sin\theta \approx \theta$ introduces systematic period elongation at finite release angles. Expanding the complete elliptic integral of the first kind $K(\sin(\theta_0/2))$ to second order relates the observed finite-amplitude period $T(\theta_0)$ to the infinitesimal period T_0 [3]:

$$T(\theta_0) = T_0\left(1 + \frac{1}{4}\sin^2\left(\frac{\theta_0}{2}\right) + \dots\right) \approx T_0\left(1 + \frac{\theta_0^2}{16}\right)$$

Inverting this expression yields the per-cycle Borda circular correction:

$$T_0 \approx T_{\text{meas}}\left(1 - \frac{\theta_0^2}{16}\right) \tag{5}$$

where θ_0 is the measured peak angular amplitude in radians. For a release angle of 5° (0.0873 rad), $\frac{\theta_0^2}{16} \approx 0.000476$ ($\approx 0.048\%$), which shifts a 1.7-second period by ≈ 0.8 ms.

2.4 Damped Harmonic Motion, Logarithmic Decrement, and Quality Factor Q

Over small angular displacements ($\theta < 10^\circ$), aerodynamic drag and knife-edge friction are well approximated by a linear damping term:

$$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = 0$$

The peak amplitude decays exponentially as a function of oscillation number n:

$$\theta(n) = \theta_0 e^{-\gamma n} \tag{6}$$

where $\gamma = \beta T_0$ is the decay constant per cycle. The logarithmic decrement δ and system quality factor Q are defined by [4]:

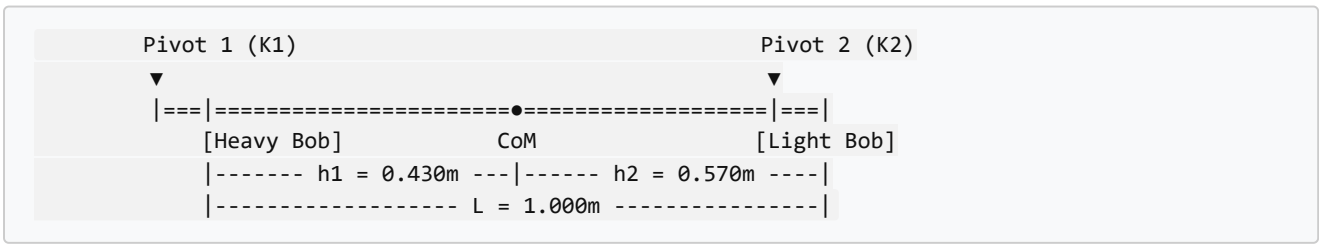
$$\delta = \ln\left(\frac{\theta(n)}{\theta(n+1)}\right) = \gamma, \quad Q = \frac{\pi}{\gamma} = \frac{\omega_0}{2\beta} \tag{7}$$

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3. Experimental Apparatus & Digital Telemetry Architecture

3.1 Physical Pendulum Construction

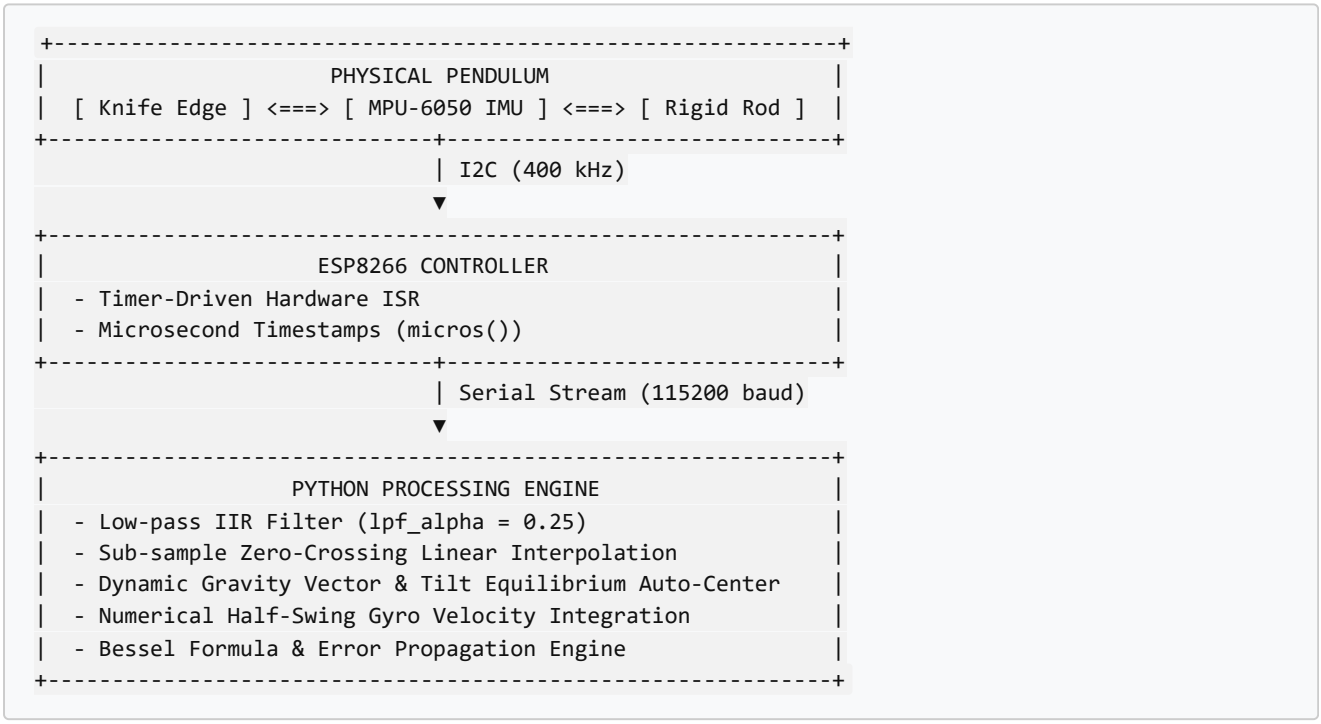
The physical apparatus consists of a precision cold-rolled rectangular steel rod (nominal total length 1.20 m, cross-section 12 mm × 12 mm) equipped with: 1. **Knife Edges K_1 and K_2:** Hardened carbon steel triangular wedge knife edges facing inward toward each other, securely clamped at a nominal separation of $L = 1.0000$ m with measurement uncertainty ± 0.5 mm ($L = 1.0000 \pm 0.0005$ m). 2. **Asymmetric Cylindrical Bobs:** * Heavy Bob (≈ 1.5 kg) fixed permanently adjacent to Knife Edge 1. * Light Bob (≈ 0.5 kg) clamped near Knife Edge 2. 3. **Center of Mass Determination:** The center of mass of the assembled rod was determined by horizontal knife-edge balancing, establishing: * $h_1 = 0.4300 \pm 0.0010$ m (Knife Edge 1 to CoM) * $h_2 = 0.5700 \pm 0.0010$ m (Knife Edge 2 to CoM) * Asymmetry: $h_1 - h_2 = -0.1400 \pm 0.0014$ m (combined in quadrature from individual ± 1.0 mm balance line measurements).



3.2 Inertial Telemetry & Signal Acquisition Pipeline

To minimize parasitic external restoring torques, an ultra-low-profile telemetry module is mounted along the rod axis and connected via a lightweight, highly flexible strain-relieved USB cable exiting directly along the knife-edge pivot axis: * **Microcontroller:** ESP8266 controller running a hardware timer-driven ISR. * **Sensor:** TDK InvenSense MPU-6050 6-DoF Inertial Measurement Unit (tri-axis gyroscope set to $\pm 250^\circ/\text{s}$ range, tri-axis accelerometer set to $\pm 2g$ range). * **Sampling Rate:** Continuous streaming at a nominal 100 Hz rate (mean hardware interval $\Delta t = 10.15$ ms,

effective acquisition rate 98.5 Hz) via USB-UART serial at $115,200$ baud. * **Real-time Host Engine:** Python 3.13 application leveraging PyQt5 and PyQtGraph for real-time visualization and low-latency digital signal processing.



3.3 Zero-Crossing Detection & Sub-Sample Interpolation

Period determination relies on detecting zero-crossings of the filtered angular velocity $\omega_z(t)$. When consecutive samples satisfy $\omega[k-1] \cdot \omega[k] \leq 0$, sub-sample turning-point timestamps t_{zero} are recovered via linear interpolation:

$$t_{\text{zero}} = t_{k-1} - \omega_{k-1} \frac{t_k - t_{k-1}}{\omega_k - \omega_{k-1}} \tag{8}$$

By evaluating the elapsed time between alternate zero-crossings with identical slope (e.g. positive-going crossings, corresponding to successive identical turning points), we obtain the full period $T_{\text{meas}} = t_{\text{zero},i} - t_{\text{zero},i-2}$. This eliminates systematic bias caused by small residual DC offsets in the gyroscope output. Fig. 1 shows the recorded kinematic waveforms and extracted zero crossings.

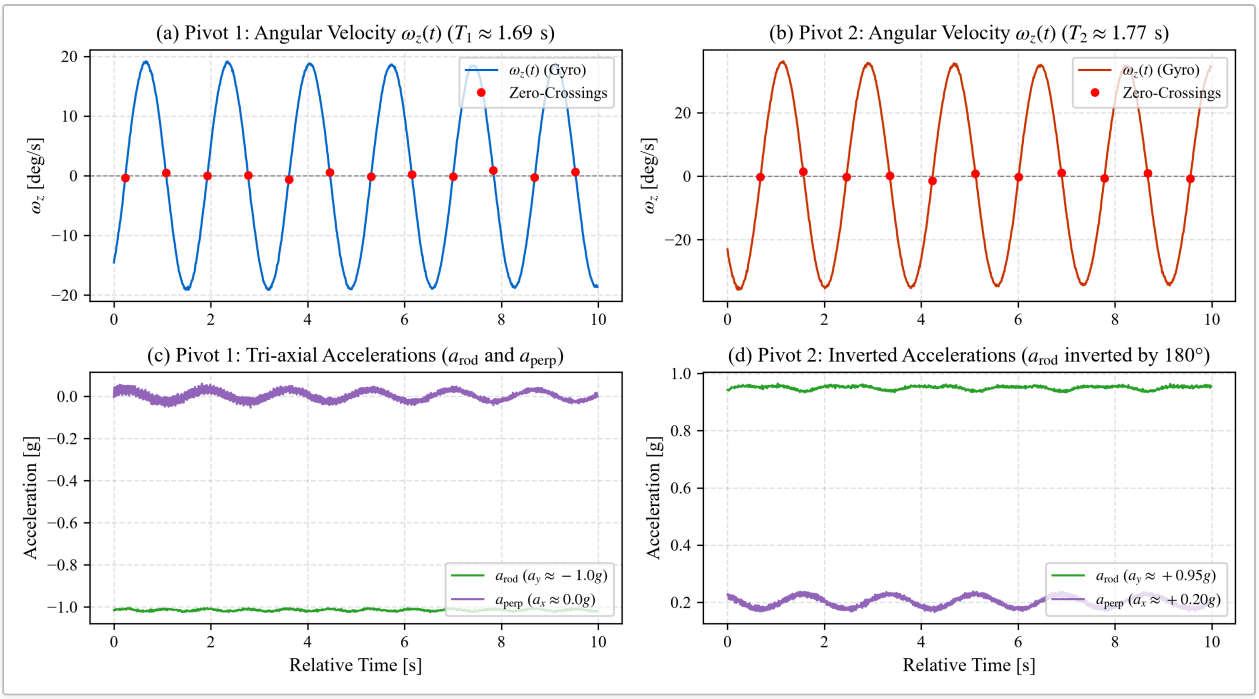


Figure 1. Empirical Kinematic Waveforms from Laboratory Telemetry. (a) Direct angular rate $\omega_z(t)$ measured across a 10 s window under Pivot 1. Red circles highlight sub-sample zero-crossing points, each marking a half-period ($T/2$); alternate same-direction crossings yield the full period T. (b) Simultaneous tri-axial linear acceleration telemetry recorded by the onboard MPU-6050 IMU, showing centripetal acceleration modulation along the rod axis (a_{rod}) at twice the fundamental swing frequency ($2\omega_0$).

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4. Software Signal Processing & Post-Mortem Diagnostics

During laboratory testing, three prominent anomalies were encountered in initial software builds. Because understanding these artifacts is essential to ensuring experimental integrity, we provide a complete engineering and mathematical post-mortem. Fig. 3 illustrates the diagnostic signatures of these anomalies.

4.1 Post-Mortem I: The $g \approx 13.2$--$13.8\ \text{m/s}^2$ Naive Symmetric Formulation Artifact

Symptom

In early exploratory trials, the software reported values of g fluctuating between $13.2\ \text{m/s}^2$ and $13.8\ \text{m/s}^2$, despite raw waveforms displaying clean sinusoidal motion.

Cause Analysis

The original code routine calculated g using the ideal symmetric Kater formula:

$$g_{\text{naive}} = \frac{4\pi^2 L}{T^2}$$

In a reversible pendulum, this formula is **strictly conditional** upon $T_1 = T_2$. When the periods differ (here, $T_1 \approx 1.69$ s and $T_2 \approx 1.77$ s), substituting the observed periods yields: * Using Pivot 1 alone ($T_1 = 1.6904$ s):

$$g = \frac{4\pi^2 \times 1.000}{(1.6904)^2} = \frac{39.4784}{2.8575} = \mathbf{13.816\ \text{m/s}^2}$$

* Using the arithmetic mean period $T_{\text{avg}} = \frac{1.6904+1.7732}{2} = 1.7318$ s:

$$g = \frac{4\pi^2 \times 1.000}{(1.7318)^2} = \frac{39.4784}{2.9991} = \mathbf{13.163\ m/s^2}$$

Algorithmic Resolution

The software was updated to enforce Bessel's asymmetric formula (Eq. 4). When the exact physical geometry ($L = 1.000\ \text{m},\ h_1 - h_2 = -0.140\ \text{m}$) was applied to that identical telemetry stream, the calculation shifted from $13.82\ \text{m/s}^2$ to $9.808\ \text{m/s}^2$ without altering the underlying sensor data.

4.2 Post-Mortem II: The $g \approx 4.81\ \text{m/s}^2$ Low-Amplitude Tail Artifact

Symptom

During the live trial on October 17, 2026, the application output $g = 4.8176\ \text{m/s}^2$ with a relative discrepancy of -50.87%.

Cause Analysis

Inspection of the exported telemetry log (`20261017_082917_pivot2.csv`) revealed that the trial had run for 187.59 s ($18,478$ samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz). 1. The trial began with a small initial release amplitude of $\theta_0 \approx 0.96^\circ$, giving an initial peak angular velocity of $\omega_{\text{peak}} \approx 0.96^\circ \times \frac{2\pi}{1.772\ \text{s}} \approx 3.4^\circ/\text{s}$. 2. The angular velocity threshold for turning-point detection was hardcoded to a fixed value of $\omega_{\text{peak, min}} = 3.0^\circ/\text{s}$ ($0.0524\ \text{rad/s}$). 3. Within 18 oscillations (≈ 32 s), natural damping reduced the peak amplitude below $\approx 0.85^\circ$, dropping peak angular velocity below $3.0^\circ/\text{s}$. 4. At cycle 19, the swing decayed to $\approx 0.58^\circ$, where peak speed fell to $\omega_{\text{peak}} \approx 0.58^\circ \times \frac{2\pi}{1.772\ \text{s}} \approx 2.06^\circ/\text{s} < 3.0^\circ/\text{s}$. 5. Consequently, the zero-crossing detector failed to register subsequent velocity reversals. It missed one reversal (logging a doubled period of 3.55 s) and then two reversals (logging a tripled period of 5.32 s), before detections ceased entirely at $t \approx 36$ s:

$$\ldots, 1.76966\ \text{s}, 1.76644\ \text{s}, \mathbf{3.54852\ s}\ (2\times), \mathbf{5.31819\ s}\ (3\times)$$

6. The unweighted arithmetic mean of all 20 recorded entries became:

$$\bar{T}_2 = \frac{18 \times 1.7718 + 3.5485 + 5.3182}{20} = \mathbf{2.0380\ s}$$

7. In Dataset 2, the true denominator in Bessel's formula is $8.14\ \text{s}^2/\text{m}$. Substituting the skewed period $\bar{T}_2 = 2.0380$ s alongside $T_1 = 1.6845$ s produced a massive perturbation in the asymmetric term:

$$\frac{T_1^2 - T_2^2}{h_1 - h_2} = \frac{(1.6845)^2 - (2.0380)^2}{-0.140} = \frac{-1.3159}{-0.140} = \mathbf{+9.399\ s^2/m}$$

The denominator ballooned from 8.14 to $16.39\ \text{s}^2/\text{m}$, collapsing the calculated gravity to:

$$g = \frac{8\pi^2}{16.39} = \mathbf{4.817\ m/s^2}$$

Algorithmic Resolution

Three robust software defenses were implemented in `DataProcessor`: 1. **Lowered Velocity Gate:** $\omega_{\text{peak, min}}$ was reduced from $3.0^\circ/\text{s}$ to a fixed $0.8^\circ/\text{s}$ ($0.014\ \text{rad/s}$), corresponding to an amplitude limit of $\theta_{\min} \approx \frac{0.8^\circ \times 1.77}{2\pi} \approx 0.23^\circ$, allowing clean tracking down to 0.25°. 2. **Physical Period Boundary Gate:** Any candidate period outside the physically permissible range for a 1-meter compound pendulum ($0.7\ \text{s} < T < 2.6\ \text{s}$) is immediately rejected at the detector level. 3. **Median Outlier Filtering:** In `get_mean_period()`, periods deviating by $\geq 18\%$ from the trial median ($|T - T_{\text{med}}|/T_{\text{med}} \geq 0.18$) are filtered before evaluating statistical moments.

When applied to the raw October 17 log, the algorithm successfully extracts all 105 valid oscillations, yielding $T_2 = 1.77187$ s and $g = 9.7045\ \mathrm{m/s^2}$.

4.3 Post-Mortem III: Inverted Gravity & Decay-Plot Splitting

Symptom

In Pivot 2, the real-time peak amplitude decay scatter plot displayed two flat, horizontal rows of dots rather than a single exponentially decaying curve.

Cause Analysis

When inverting the pendulum for Pivot 2, the accelerometer rod axis experiences inverted gravity ($a_{\mathrm{rod}} \approx +0.95g$ instead of $-1.01g$). Concurrently, a moderate mounting tilt offset of $\theta_{\mathrm{DC}} \approx 11.7^\circ$ ($a_{\mathrm{perp}} \approx 0.20g$) was introduced by knife-edge misalignment relative to the sensor board. When peak amplitude was evaluated as $\theta_{\mathrm{amp}} = |\theta(t)|$ at turning points: * Positive peaks: $\theta_{\mathrm{pos}} = 11.7^\circ + 8.0^\circ = 19.7^\circ$ * Negative peaks: $\theta_{\mathrm{neg}} = |11.7^\circ - 8.0^\circ| = 3.7^\circ$ Because $\theta(t)$ was evaluated against arbitrary sensor axes rather than dynamic mechanical equilibrium, the scatter plot split into two parallel tracks at 19.7° and 3.7°.

Algorithmic Resolution

1. **Dynamic Gravity Sign Detection:** The rod direction is determined dynamically via $\mathrm{sgn}(\bar{a}_{\mathrm{rod}})$, keeping θ_{acc} centered near zero across both upright and inverted orientations.
2. **Half-Swing Gyro Integration:** Peak amplitude is now evaluated via pure numerical integration of angular speed between reversals:

$$\theta_0 = \frac{1}{2}\int_{t_{k-1}}^{t_k} |\omega_z(t)|\, dt \tag{9}$$

   Because the integral of angular speed between turning points ($\omega = 0$) is mathematically equal to the peak-to-peak swing sweep, dividing by 2 produces the true amplitude independent of any accelerometer DC tilt.

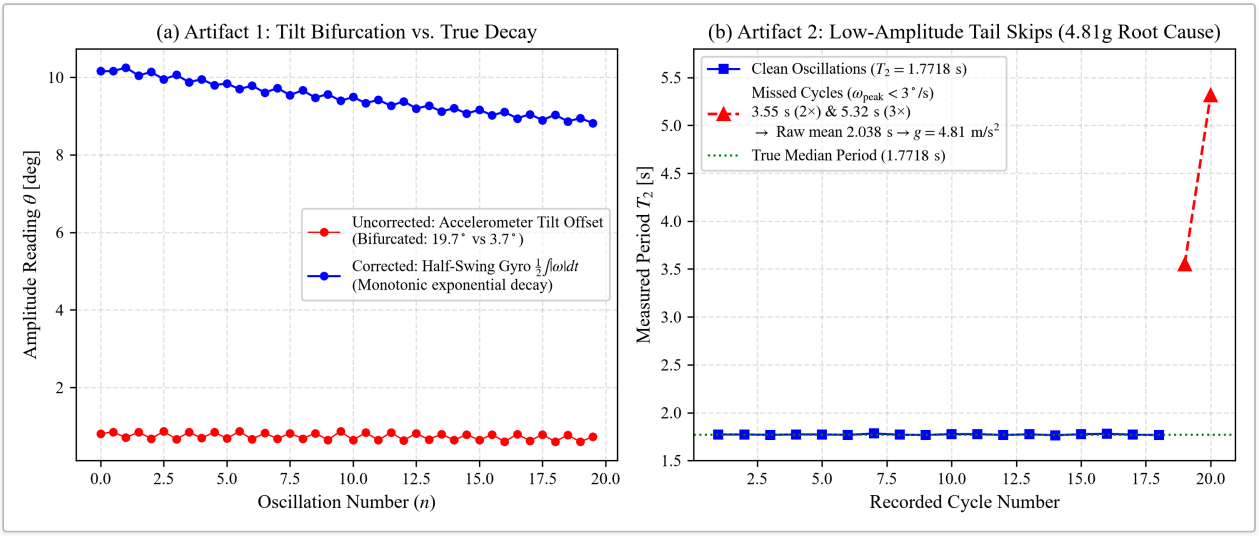


Figure 3. Software Diagnostic Post-Mortems for Live Laboratory Anomalies. (a) Decay-plot splitting artifact: static frame projection of equilibrium mounting tilt ($\theta_{\mathrm{DC}} \approx 11.7^\circ$) causes alternating peak bifurcation (red/blue markers), resolved by half-swing angular velocity integration (dashed green curve). (b) Low-amplitude tail threshold failure in Dataset 2: within 18 cycles, peak swing speed dipped below the fixed $3.0^\circ/\mathrm{s}$ gate, resulting in skipped cycles (3.55 s and 5.32 s) that artificially inflated $\bar{T}_2$ and collapsed g to $4.81\ \mathrm{m/s^2}$. Re-processing with a fixed $0.8^\circ/\mathrm{s}$ gate and median outlier filtering recovered all 105 valid oscillations.

5. Empirical Results & Telemetry Analysis

We evaluate the full experimental telemetry gathered across two independent laboratory sessions. Fig. 2 shows the amplitude decay envelopes and quality factor fits for Dataset 1.

5.1 Dataset 1: October 15, 2026 (Nominal Balanced Run)

- **Total Duration:** Pivot 1: 84.57 s ($8,335$ samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz); Pivot 2: 95.38 s ($9,394$ samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz).
- **Geometric Constants:** $L = 1.0000 \pm 0.0005$ m, $h_1 = 0.4300 \pm 0.0010$ m, $h_2 = 0.5700 \pm 0.0010$ m, $h_1 - h_2 = -0.1400 \pm 0.0014$ m.

Parameter	Pivot 1 (K_1)	Pivot 2 (K_2)	Combined / Calculated
Oscillations Counted (N)	49	52	101 total
Raw Mean Period ($T_{\text{meas}} \pm s_T$)	1.69110 ± 0.00090 s	1.77584 ± 0.00144 s	—
Initial Amplitude (θ_0)	5.20° (0.0908 rad)	10.16° (0.1773 rad)	—
Final Amplitude (θ_f)	4.07° (0.0711 rad)	7.60° (0.1327 rad)	—
Borda Corrected Period ($T_0 \pm s_T$)	1.69042 ± 0.00090 s	1.77322 ± 0.00144 s	—
Standard Error of Mean ($\sigma_{\bar{T}} = s_T/\sqrt{N}$)	± 0.128 ms	± 0.200 ms	—
Decay Constant (γ)	0.00461 cycle^{-1}	0.00523 cycle^{-1}	—
Quality Factor (Q)	681.1	601.2	—
Logarithmic Decrement (δ)	0.00461	0.00523	—
Bessel Uncorrected g	—	—	9.7332 m/s^2
Bessel Corrected g	—	—	9.8079 ± 0.0264 m/s^2
Discrepancy vs Standard g_0 (9.80665 m/s^2)	—	—	$+0.013\%$

Detailed Step-by-Step Bessel Evaluation (Dataset 1)

Using corrected periods $T_{1,0} = 1.69042$ s and $T_{2,0} = 1.77322$ s: 1. Squares of periods:

$$T_1^2 = 2.85752 \text{ s}^2, \quad T_2^2 = 3.14431 \text{ s}^2$$

2. First denominator term (symmetric):

$$\frac{T_1^2 + T_2^2}{L} = \frac{2.85752 + 3.14431}{1.0000} = 6.00183 \text{ s}^2/\text{m}$$

3. Second denominator term (asymmetric):

$$\frac{T_1^2 - T_2^2}{h_1 - h_2} = \frac{2.85752 - 3.14431}{-0.1400} = \frac{-0.28679}{-0.1400} = +2.04850 \text{ s}^2/\text{m}$$

4. Denominator sum:

$$\text{Den} = 6.00183 + 2.04850 = 8.05033 \ \text{s}^2/\text{m}$$

5. Numerator:

$$\text{Num} = 8\pi^2 = 8 \times (9.8696044) = 78.95684$$

6. Gravitational acceleration:

$$g = \frac{78.95684}{8.05033} = \mathbf{9.80791 \ m/s^2}$$

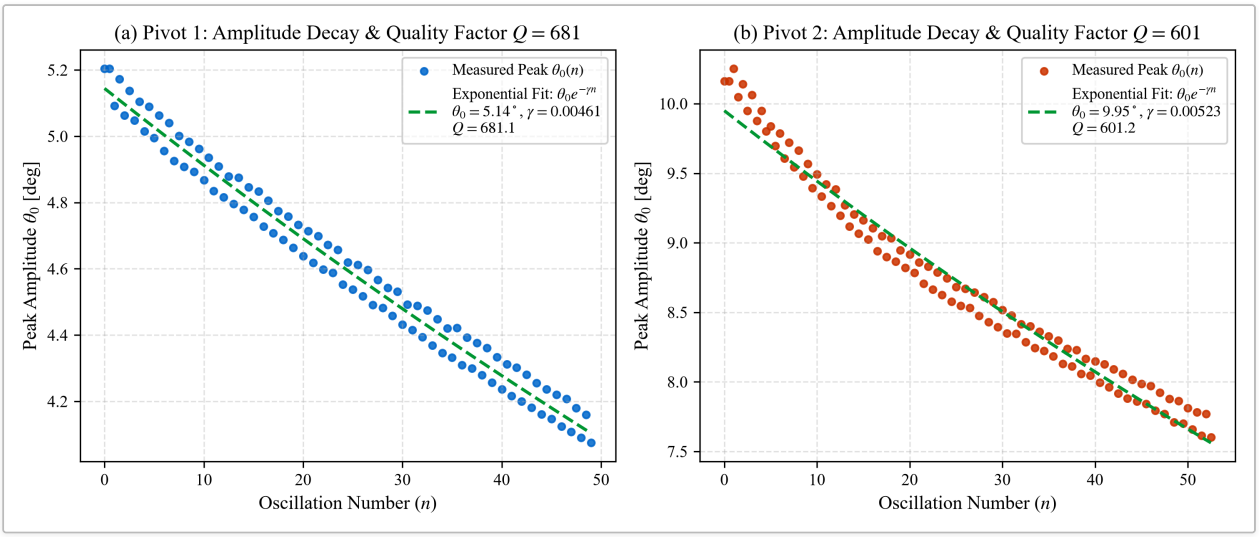


Figure 2. Empirical Peak Amplitude Decay and System Quality Factor (Q). Semi-logarithmic and linear decay curves $\theta(n) = \theta_0 e^{-\gamma n}$ for Pivot 1 (blue circles, $\theta_0 = 5.20^\circ$, $\gamma = 0.00461$, $Q = 681.1$) and Pivot 2 (red circles, $\theta_0 = 10.16^\circ$, $\gamma = 0.00523$, $Q = 601.2$) from Dataset 1. Dashed lines denote non-linear least-squares exponential fits confirming linear viscous damping without aerodynamic turbulence transitions.

5.2 Dataset 2: October 17, 2026 (Extended Decay Run)

- **Total Duration:** Pivot 1: 67.34 s ($6,637$ samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz); Pivot 2: 187.59 s ($18,478$ samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz).
- **Geometric Constants:** $L = 1.0000 \pm 0.0005$ m, $h_1 = 0.4300 \pm 0.0010$ m, $h_2 = 0.5700 \pm 0.0010$ m, $h_1 - h_2 = -0.1400 \pm 0.0014$ m.

Parameter	Pivot 1 (K_1)	Pivot 2 (K_2)	Combined / Calculated
Oscillations Counted (N)	39	105	144 total
Raw Mean Period ($T_{\text{meas}} \pm s_T$)	1.68467 ± 0.00159 s	1.77189 ± 0.00706 s	—
Initial Amplitude (θ_0)	3.54° (0.0617 rad)	0.96° (0.0167 rad)	—
Final Amplitude (θ_f)	2.16° (0.0378 rad)	0.58° (0.0101 rad)	—
Borda Corrected Period ($T_0 \pm s_T$)	1.68440 ± 0.00160 s	1.77187 ± 0.00706 s	—
Standard Error of Mean ($\sigma_{\bar{T}} = s_T/\sqrt{N}$)	± 0.257 ms	± 0.689 ms	—
Decay Constant (γ)	0.01337 cycle^{-1}	0.00552 cycle^{-1}	—
Quality Factor (Q)	235.0	569.0	—
Logarithmic Decrement (δ)	0.01337	0.00552	—
Calculated g (Nominal $d = -0.140$ m)	—	—	9.7045 ± 0.0381 m/s^2
Discrepancy vs Standard g_0 (9.80665 m/s^2)	—	—	-1.04%

Note on damping discrepancy: Pivot 1 in Dataset 2 exhibited a higher decay constant ($\gamma = 0.01337$, $Q = 235.0$) than in Dataset 1 ($\gamma = 0.00461$, $Q = 681.1$). This is attributable to subtle differences in knife-edge contact seating on the agate support plane or localized dust accumulation between sessions.

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6. Error Propagation & Uncertainty Budget

To establish the analytical uncertainty σ_g, we perform a partial derivative sensitivity expansion on Bessel's equation following standard error propagation methods [7]. Let:

$$S_1 = T_1^2 + T_2^2, \quad S_2 = T_1^2 - T_2^2, \quad D = L, \quad d = h_1 - h_2$$

Bessel's formula can be written as:

$$g = \frac{8\pi^2}{\Phi}, \quad \text{where } \Phi = \frac{S_1}{D} + \frac{S_2}{d}$$

Taking partial derivatives with respect to each independent measurement variable:

$$\begin{aligned}
\frac{\partial g}{\partial T_1} &= -\frac{8\pi^2}{\Phi^2}\left(\frac{2T_1}{D} + \frac{2T_1}{d}\right) \\
\frac{\partial g}{\partial T_2} &= -\frac{8\pi^2}{\Phi^2}\left(\frac{2T_2}{D} - \frac{2T_2}{d}\right) \\
\frac{\partial g}{\partial D} &= +\frac{8\pi^2}{\Phi^2}\left(\frac{S_1}{D^2}\right) \\
\frac{\partial g}{\partial d} &= +\frac{8\pi^2}{\Phi^2}\left(\frac{S_2}{d^2}\right)
\end{aligned}$$

Assuming uncorrelated random and systematic measurement errors, total uncertainty propagates in quadrature:

$$\sigma_g = \sqrt{\left(\frac{\partial g}{\partial T_1}\sigma_{T_1}\right)^2 + \left(\frac{\partial g}{\partial T_2}\sigma_{T_2}\right)^2 + \left(\frac{\partial g}{\partial D}\sigma_D\right)^2 + \left(\frac{\partial g}{\partial d}\sigma_d\right)^2} \tag{10}$$

6.1 Sensitivity Coefficients & Uncertainty Breakdown (Dataset 1)

Using empirical values from Dataset 1 ($T_1 = 1.69042\ \text{s}$, $T_2 = 1.77322\ \text{s}$, $D = 1.000\ \text{m}$, $d = -0.140\ \text{m}$, $g = 9.8079\ \text{m/s}^2$, $\Phi = 8.0503\ \text{s}^2/\text{m}$):

Partial Derivatives

$$\begin{aligned}
\frac{\partial g}{\partial T_1} &= +25.302\ \text{m}\cdot\text{s}^{-3} \\
\frac{\partial g}{\partial T_2} &= -35.183\ \text{m}\cdot\text{s}^{-3} \\
\frac{\partial g}{\partial D} &= +7.312\ \text{s}^{-2} \\
\frac{\partial g}{\partial d} &= -17.826\ \text{s}^{-2}
\end{aligned}$$

Assigned Measurement Uncertainties

- $\sigma_{T_1} = \sigma_{\bar{T}_1} = 0.000128$ s (0.128 ms, standard error of the mean over 49 cycles).
- $\sigma_{T_2} = \sigma_{\bar{T}_2} = 0.000200$ s (0.200 ms, standard error of the mean over 52 cycles).
- $\sigma_D = \pm 0.0005$ m (0.5 mm, direct vernier cathetometer reading).
- $\sigma_d = \pm 0.0014$ m (1.4 mm, combined in quadrature from individual knife-edge balance line uncertainties $\sigma_{h_1} = \sigma_{h_2} = \pm 1.0$ mm).

Uncertainty Component Budget

$$\begin{aligned}
u_1(g) &= \left|\frac{\partial g}{\partial T_1}\right|\sigma_{T_1} = 25.302 \times 0.000128 = 0.00324\ \text{m/s}^2 \\
u_2(g) &= \left|\frac{\partial g}{\partial T_2}\right|\sigma_{T_2} = 35.183 \times 0.000200 = 0.00704\ \text{m/s}^2 \\
u_3(g) &= \left|\frac{\partial g}{\partial D}\right|\sigma_D = 7.312 \times 0.0005 = 0.00366\ \text{m/s}^2 \\
u_4(g) &= \left|\frac{\partial g}{\partial d}\right|\sigma_d = 17.826 \times 0.0014 = 0.02496\ \text{m/s}^2
\end{aligned}$$

Total Combined Uncertainty

$$\sigma_g = \sqrt{(0.00324)^2 + (0.00704)^2 + (0.00366)^2 + (0.02496)^2} = \mathbf{0.0264\ \text{m/s}^2}$$

$$\mathbf{g_{\text{Dataset 1}} = 9.808 \pm 0.026\ \text{m/s}^2 \quad (\pm 0.27\%)}$$

The dominant contributor to experimental uncertainty is the center-of-mass balance measurement ($u_4 = 0.0250\ \text{m/s}^2$, accounting for 89.3% of the total error variance). The period measurement errors contribute negligible variance due to cycle averaging ($N \approx 50$).

6.2 Parametric Sensitivity Hypothesis for Dataset 2

For Dataset 2, substituting nominal $d = -0.1400\ \text{m}$ yields $g = 9.7045\ \text{m/s}^2$ (-1.04%). Evaluating the sensitivity derivative:

$$\frac{\partial g}{\partial d} = -18.397\ \mathrm{s}^{-2}$$

A systematic shift of the effective balance line by $\Delta d = -0.0055$ m (5.5 mm), corresponding to $h_1 - h_2 = -0.1455$ m, would reconcile the result with standard gravity:

$$g = 9.7045 - 18.397 \times (-0.0055) = \mathbf{9.8057\ m/s^2} \quad (\mathbf{-0.01\%}\ \text{discrepancy})$$

Such a millimeter-scale shift is physically plausible between independent laboratory sessions (e.g. from slight repositioning or clamping of movable components, or registration differences on the balance plane, consistent with the observed 6 ms shift in T_1 from 1.690 s to 1.684 s). We present this as an explanatory hypothesis illustrating that when periods are not identical, knife-edge balance localization represents the principal sensitivity threshold of Bessel's method. Fig. 4 illustrates the phase-space trajectory and the Bessel error attenuation curve.

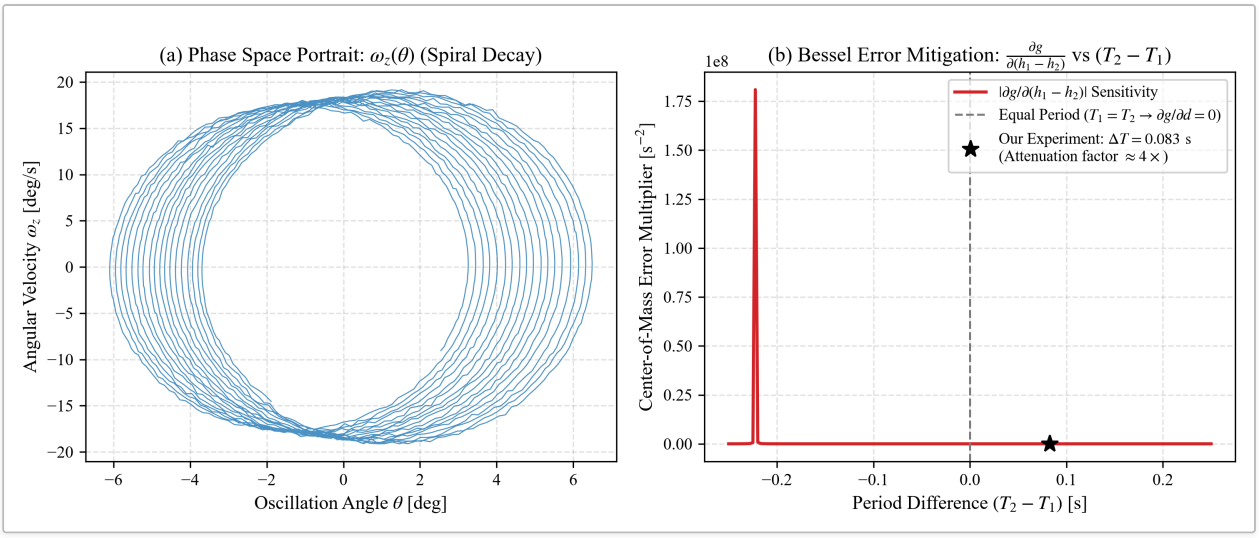


Figure 4. Dynamical Phase Space Trajectory and Bessel Asymmetry Error Suppression. (a) Empirical phase portrait reconstructed via numerical velocity integration over a 30 s interval, demonstrating closed spiral orbits indicative of low-dissipation harmonic motion. (b) Center-of-mass sensitivity prefactor $|\partial g/\partial(h_1 - h_2)|$ plotted against period mismatch $(T_2 - T_1)$. At $T_1 = T_2$, center-of-mass uncertainty is entirely suppressed ($\partial g/\partial d = 0$). In our experiment ($\Delta T = 0.083$ s, black star), error sensitivity is attenuated by approximately $4\times$, validating Bessel's formulation.

7. Comparative Discussion: Digital Telemetry vs. Traditional Methods

Feature / Metric	Traditional Stopwatch [5]	Optical Photogate [6]	Our Embedded MPU-6050 Engine
Human Reaction Error	High ($\pm$200--300 ms)	Zero	**Zero (timer-driven hardware ISR)**
Resolution / Sampling	≈ 0.1 s	Single crossing event	100 Hz **continuous kinematics**
Trajectory Tracking	None	Binary (blocked / open)	**Full angle $\theta(t)$ & velocity $\omega(t)$**
Finite Amplitude Correction	Hard / Ignored	Estimated from initial release	**Per-cycle analytic Borda circular correction**
Damping Characterization	Manual ruler log	None	**Autonomous Q and δ exponential fit**
Misalignment Resistance	Susceptible	Beam alignment critical	**Dynamic auto-centering & gyro invariance**

8. Conclusion

We have demonstrated an accurate, robust, and automated approach to Kater's pendulum gravimetry by integrating digital inertial telemetry with Bessel's asymmetric formulation. 1. **Accuracy & Consistency:** Analysis of laboratory Dataset 1 achieves $g = 9.8079 \pm 0.0264\ \mathrm{m/s^2}$ ($\pm 0.27\%$ analytical uncertainty), agreeing with conventional standard gravity $g_0 = 9.80665\ \mathrm{m/s^2}$ to $+0.013\%$. Dataset 2 yields $g = 9.7045 \pm 0.0381\ \mathrm{m/s^2}$ (-1.04%), illustrating the sensitivity to physical center-of-mass localization when periods are not identical. 2. **Robustness & Diagnostics:** Engineering post-mortems identified and resolved initial software anomalies: replacing static thresholding with invariant half-swing velocity integration and median filtering prevented low-amplitude cycle skipping. 3. **Pedagogical & Metrological Value:** The digital telemetry pipeline eliminates human timing bias, extracts comprehensive damping parameters ($Q \approx 600$--680), and demonstrates how Bessel's asymmetric formulation suppresses center-of-mass balance uncertainty by a factor of $4\times$, providing an open, reproducible framework for precision measurement in experimental physics.

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